Physical and Organic Chemistry

Science

May, 2026

South East Technological University
Ollscoil Teicneolaíochta an Oirdheiscirt

Physical and Organic Chemistry (A07611)

Short Title:	Physical and Organic Chemistry		
Faculty	Science and Computing		
Department	Science		
Cluster	Chemistry		
Author	EMLANDERS		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This module provides an introduction to the general principles of physical and organic chemistry and their applications. Topics include thermochemistry, chemical equilibrium, acid/base reactions, pH scale, kinetics and catalysis. In addition, an appreciation for the scope of organic chemistry, including hydrocarbons, homologous series, isomerism, functional groups and fundamental organic reactions will be obtained.

Programmes

`CHEM-0002`	BSc (Common Entry) (`WD_SSCIE_D`)	1 / 2 / M
`CHEM-0002`	BSc (Hons) in Agricultural Science (`WD_SAGRI_B`)	1 / 2 / M
`CHEM-0002`	BSc (Hons) in Food Science and Innovation (`WD_SFSIN_B`)	1 / 2 / M
`CHEM-0002`	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (`WD_SMBIO_B`)	1 / 2 / M
`CHEM-0002`	BSc (Hons) in Pharmaceutical Science (`WD_SPHSC_B`)	1 / 2 / M
`CHEM-0002`	BSc (Hons) in Physics for Modern Technology (`WD_KPHTE_B`)	1 / 2 / M
`CHEM-0002`	BSc (Hons) in Science (Common Entry) (`WD_SSCCM_B`)	1 / 2 / M
`CHEM-0002`	BSc in Food Science with Business (`WD_SFDSC_D`)	1 / 2 / M
`CHEM-0002`	BSc in Molecular Biology with Biopharmaceutical Science (`WD_SMBIO_D`)	1 / 2 / M
`CHEM-0002`	BSc in Pharmaceutical Science (`WD_SPHSC_D`)	1 / 2 / M

Indicative Content

- Thermochemistry: heats of formation, combustion, neutralisation and solution; energy changes during chemical reactions
- Equilibrium in physical processes
- Chemical equilibrium, equilibrium constants & Le Chatelier's Principle
- Ionic equilibria in aqueous solution and the ionic product of water; pH scale and the buffer equation
- Arrhenius equation and the Arrhenius concept of acids and bases
- Representation of structural formulae and nomenclature of organic molecules
- Hydrocarbons, homologous series: bonding in alkanes, cycloalkanes, alkenes, alkynes and aromatics; isomerism
- Functional groups
- Nucleophiles and electrophiles; reactions of alkyl halides: Sn1, Sn2, E1, E2 and electrophilic addition reactions

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Calculate heats of formation, combustion, neutralisation, solution and energy changes during a chemical reaction.
2. Outline the principles of chemical equilibrium and solve related problems: pH, Ka, Kb.
3. Apply the Henderson-Hasselbalch equation in order to determine pH.
4. Name and draw organic molecules.
5. Identify nucleophiles, electrophiles and functional groups.
6. Describe reaction mechanisms: Sn1, Sn2, E1, E2 and electrophilic addition reactions.
7. Complete functional group testing, basic organic synthesis, recrystallisation, conductivity/pH/colorimetry experiments.
8. Demonstrate a range of laboratory skills and safe work practices in the physical and organic chemistry laboratory environments.

Learning and Teaching Methods

- Lectures.
- Tutorials.
- Laboratory-based practicals.
- Self-directed study.

Assessment Methods

	Weighing		Outcomes Assessed
Final Written Examination	40%		1-6
Continuous Assessment	60%		
Practical		40%	7,8
Group Project		20%	1-6

Assessment Criteria

<40%: No understanding of the basic principles of physical and organic chemistry. Inability to carry out practical skills in the associated laboratories.

40%-49%: Has a superficial knowledge of physical and organic chemistry. Has a basic level of competency & skills in the associated laboratories.

50%-59%: Shows a reasonable level of understanding of the various principles involved in physical and organic chemistry. Displays good practical skills in the associated laboratories.

60%-69%: Shows a good knowledge of the areas of physical and organic chemistry, as well as an ability to reason and analyse problems related to the area. Has a high level of skills in the associated laboratories.

70%-100%: Excellent knowledge of the areas of physical and organic chemistry. Excellent reasoning skills and focused well thought out arguments. Engages in debate in class and has a high level of efficiency in the associated laboratories.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Tutorial	8	
Practical	16	
Independent Learning	75	

Essential Material

- "mastering chemistry." www.pearsonmylabandmastering.com/
- Browne, T.L. and H.E. LeMay. _Chemistry: The Central Science_. 3rd ed. Australia: Pearson Australia, 2013.

Supplementary Material

- Atkins, P.W. and J. De Paulo. _Physical Chemistry_. 9th ed. Oxford: Oxford University Press, 2009.
- Ebbing, D. and S. Gammon. _General Chemistry_. 11th ed. New York: Cengage learning, 2016.
- Russo, S. and M. Silver. _Introductory Chemistry: Atoms First_. 5th ed. San Francisco: Prentice hall, 2010.
- Silberberg , M.S. and P.G. Amateis. _Chemistry, The Molecular Nature of Matter and Change_. 7th ed. UK: McGraw-Hill , 2015.

Requested Resources

- Room Type: Chemistry Lab